

```
> b <- 3

> sum(a + b)
Error in a + b : non-numeric argument to binary operator
```

Unexpected symbol

The function arguments need to be separated with a comma. If the comma is missed, an 'unexpected symbol' error is returned, as shown below:

```
> cut(mtcars$hp breaks=2)
Error: unexpected symbol in "cut(mtcars$hp breaks"

> cut(mtcars$hp,breaks=2)
 [1] (51.7,194] (51.7,194] (51.7,194] (51.7,194] (51.7,194]
(51.7,194]
 [7] (194,335] (51.7,194] (51.7,194] (51.7,194] (51.7,194]
(51.7,194]
[13] (51.7,194] (51.7,194] (194,335] (194,335] (194,335]
(51.7,194]
[19] (51.7,194] (51.7,194] (51.7,194] (51.7,194] (51.7,194]
(194,335]
[25] (51.7,194] (51.7,194] (51.7,194] (51.7,194] (194,335]
(51.7,194]
[31] (194,335] (51.7,194]
Levels: (51.7,194] (194,335]
```

Unexpected numeric constant

The variable assignment needs to use the <- construct, as illustrated below:

```
> pi 3.14
Error: unexpected numeric constant in "pi 3.14"

> pi <- 3.14
> pi
[1] 3.14
```

Operator is invalid for atomic vectors

The R operations are defined for specific data structures. We cannot use the data frame operator (\$) with vectors. For example:

```
> a <- c(1, 2, 3)

> a$foo
Error in a$foo : $ operator is invalid for atomic vectors
```

Object cannot be coerced to double

An R object can contain lists, and any direct operations on the same will result in a coercion error.

```
> n <- list(1:3, 5:7)
> n
[[1]]
[1] 1 2 3

[[2]]
[1] 5 6 7

> as.numeric(n)
Error: 'list' object cannot be coerced to type 'double'
```

You need to use the *unlist()* function to act on the individual elements, as shown below:

```
> as.numeric(unlist(n))
[1] 1 2 3 5 6 7
```

Attempt to apply non-function

An operator needs to be used between R objects; otherwise the 'attempt to apply non-function' error is thrown by R, as illustrated below:

```
> a <- c(1, 2, 3)
> b <- c(4, 5, 6)

> (a)(b)
Error: attempt to apply non-function
> (a) * (b)
[1] 4 10 18
```

Missing value where true/false is needed

A data set can contain missing values, labelled as 'NA'. The *is.na()* function needs to be used to check for such values. Otherwise, a 'missing value' error is reported by R.

```
> x = NA

> if (x) { print('Missing data') }
Error in if (x) { : missing value where TRUE/FALSE needed

> if (is.na(x)) { print('Missing data') }
[1] "Missing data"
```

Plotting errors

The UCI Machine Learning Repository contains a 'Bank Marketing Data Set' for a Portuguese banking institution available at <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>. The data can be used for public research use. There are four data sets available, and we will use the *read.csv()* function to import the data from a 'bank.csv' file into a data frame.